

Advanced (Habanero)

- Q1.** Solve for x in $x^2 + 5x + 6 = 0$ using the quadratic formula.
(A) $x = -2, -3$
(B) $x = -1, -6$
(C) $x = 2, 3$
(D) $x = 1, 6$
(E) $x = -2, 3$
- Q2.** What is the discriminant of $x^2 - 4x - 5 = 0$?
(A) -24
(B) -16
(C) 4
(D) 16
(E) 24
- Q3.** How many solutions does $x^2 + 2x + 1 = 0$ have?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q4.** Solve for x in $x^2 - 7x + 12 = 0$ using the quadratic formula.
(A) $x = 3, 4$
(B) $x = 2, 6$
(C) $x = 1, 7$
(D) $x = 3, 5$
(E) $x = 2, 5$
- Q5.** What is the discriminant of $x^2 - 3x - 2 = 0$?
(A) -5
(B) -1
(C) 5
(D) 13
(E) 17
- Q6.** How many solutions does $x^2 + 5x + 6 = 0$ have?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q7.** Solve for x in $x^2 - 2x - 8 = 0$ using the quadratic formula.
(A) $x = -2, 4$
(B) $x = 1, 8$
(C) $x = -4, 2$
(D) $x = 2, 8$
(E) $x = 4, -2$
- Q8.** What is the discriminant of $x^2 + 2x + 2 = 0$?
(A) -4
(B) 0
(C) 2
(D) 4
(E) 8

Open Ended Questions (FRQ)

- Q9.** Solve $x^2 + 4x + 4 = 0$ using the quadratic formula and interpret the discriminant. Give the number and type of solutions.
- Q10.** Given $ax^2 + bx + c = 0$, derive the quadratic formula and explain how the discriminant $b^2 - 4ac$ determines the number and type of solutions.

Chef's Tips

- Chef's Tips
- Remind students to apply the quadratic formula correctly
- Check student understanding of the relationship between the discriminant and the number and type of solutions